

Utility Theory and Decision Making Under Uncertainty

Preferences, Lotteries, Expected Utility and Strategic Behaviour

Call or Fold ?

SciMathSoc

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Outline

- 1 Utility Theory
- 2 Lotteries
- 3 Risk Attitudes
- 4 Discounted Payoffs
- 5 Allais Paradox
- 6 Repeated Prisoner's Dilemma

A Simple Decision Problem

You must choose between:

Option A

Receive Rs. 100 with certainty.

Option B

50% chance of Rs. 250

50% chance of Rs. 0

Questions:

- Which option would you choose?
- Why?
- Can such choices be analyzed mathematically?

Why Utility Theory?

Utility theory provides a mathematical framework for:

- Comparing alternatives
- Modeling rational decision making
- Analyzing choices under uncertainty
- Understanding risk-taking behavior
- Studying strategic interactions

Central Question

How can preferences be represented and analyzed mathematically?

Roadmap

- 1 Utility Theory
- 2 Lotteries
- 3 Risk Attitudes
- 4 Discounted Payoffs
- 5 Allais Paradox
- 6 Repeated Prisoner's Dilemma

Preference Relations

Let X denote the set of alternatives.

A preference relation is denoted by

$$\succeq$$

where

$$x \succeq y$$

means

" x is at least as preferred as y ."

Derived relations:

$$x \succ y$$

Strict Preference

Axiom 1: Reflexivity

A preference relation is reflexive if

$$x \succeq x$$

for every alternative x .

Interpretation

Every option is at least as good as itself.

Examples:

- A book compared to itself
- A lottery compared to itself
- A strategy compared to itself

Axiom 2: Completeness

For any two alternatives x and y ,

$$x \succeq y$$

or

$$y \succeq x$$

(or both).

Interpretation

Every pair of alternatives can be compared.

Example:

Tea vs Coffee

The individual must be able to rank them.

Axiom 3: Transitivity

A preference relation is transitive if

$$x \succeq y$$

and

$$y \succeq z$$

imply

$$x \succeq z$$

Interpretation

If A is preferred to B and B is preferred to C, then A should be preferred to C.

Why Is Transitivity Important?

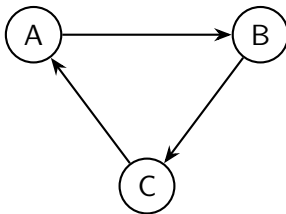
Without transitivity we may obtain:

$$A \succ B$$

$$B \succ C$$

$$C \succ A$$

This creates a preference cycle.



Rational Preferences

A preference relation is rational if it satisfies:

- 1 Reflexivity
- 2 Completeness
- 3 Transitivity

Fundamental Result

Every rational preference relation can be represented by a utility function.

This motivates Utility Theory.

From Preferences to Utility

Preferences tell us how alternatives are ranked.

Utility functions assign numbers to these rankings.

$$u : X \rightarrow \mathbb{R}$$

such that

$$x \succeq y \iff u(x) \geq u(y)$$

Key Idea

Utility does not measure happiness directly.

It provides a numerical representation of preferences.

Utility Representation Example

Suppose a student has preferences:

$$Pizza \succ Burger \succ Sandwich$$

One possible utility representation is:

Alternative	Utility
Pizza	10
Burger	7
Sandwich	3

The ranking is preserved:

$$10 > 7 > 3$$

Ordinal Utility

Ordinal utility only preserves rankings.

If

$$u(Pizza) = 10$$

and

$$u(Burger) = 5$$

we only know that Pizza is preferred.

Important

The difference

$$10 - 5$$

has no meaningful interpretation.

Monotonic Transformations

Suppose $u=(10,7,3)$ represents preferences.

Then, $v(x)=2u(x)+5$ also represents the same preferences.

Alternative	$u(x)$	$v(x)$
Pizza	10	25
Burger	7	19
Sandwich	3	11

Any strictly increasing transformation preserves preference ordering.

Utility Representation Theorem

Theorem

If a preference relation is:

- Complete
- Reflexive
- Transitive

then there exists a utility function

$$u : X \rightarrow \mathbb{R}$$

that represents those preferences.

This theorem provides the bridge between qualitative preferences and quantitative analysis.

Decision Making Under Uncertainty

So far, we have studied preferences over certain outcomes.

Examples:

- Pizza vs Burger
- Tea vs Coffee
- Strategy A vs Strategy B

In reality, outcomes are often uncertain.

Examples

Investments, Insurance, Gambling, Research Projects

To model uncertainty, economists use **lotteries**.

What is a Lottery?

A lottery is a probability distribution over outcomes.

$$L = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}$$

where

$$p_i \geq 0, \quad \sum p_i = 1$$

Interpretation

A lottery specifies every possible outcome and its probability.

Anatomy of a Lottery

$$L = \begin{pmatrix} 100 & 50 & 0 \\ 0.5 & 0.3 & 0.2 \end{pmatrix}$$

Outcome	Probability	Contribution
100	0.5	50
50	0.3	15
0	0.2	0

$$EV = 65$$

Simple Lottery Example 1

Coin Toss Gamble

$$L = \begin{pmatrix} 100 & 0 \\ 0.5 & 0.5 \end{pmatrix}$$

Outcome	Probability
Rs.100	0.5
Rs.0	0.5

$$EV = 50$$

Simple Lottery Example 2

Scholarship Allocation

Award	Probability
Rs.50,000	0.1
Rs.20,000	0.3
Rs.0	0.6

$$EV = 11000$$

Question

Would you accept Rs.11,000 for certain instead?

Simple Lottery Example 3

Rolling a Fair Die

$$L = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

$$EV = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Observation

Expected value need not be a possible outcome.

Compound Lottery: Definition

A compound lottery is a lottery whose outcomes are themselves lotteries.

Formally, instead of assigning probabilities directly to outcomes,

$$L = (L_1, p_1; L_2, p_2; \dots; L_n, p_n)$$

where

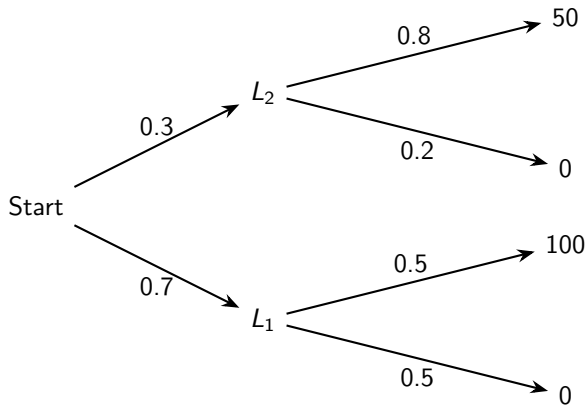
$$\sum_{i=1}^n p_i = 1$$

and each L_i is itself a lottery.

Interpretation

A compound lottery involves multiple stages of uncertainty.

Tree Representation of a Compound Lottery



Interpretation

Stage 1 determines which lottery is selected. Stage 2 determines the final monetary outcome.

Reduction of Compound Lotteries

For outcome Rs.100:

$$P(100) = 0.7 \times 0.5 = 0.35$$

For outcome Rs.50:

$$P(50) = 0.3 \times 0.8 = 0.24$$

For outcome Rs.0:

$$P(0) = 0.7(0.5) + 0.3(0.2) = 0.41$$

Equivalent Simple Lottery

The compound lottery

$$(L_1, 0.7; L_2, 0.3)$$

is equivalent to

$$\begin{pmatrix} 100 & 50 & 0 \\ 0.35 & 0.24 & 0.41 \end{pmatrix}$$

Reduction Principle

Every compound lottery can be represented as an equivalent simple lottery.

Why Is Reduction Important?

Expected Utility Theory evaluates lotteries.

Without reduction, compound lotteries would require a separate theory.

The Reduction Principle allows us to:

- Treat all lotteries uniformly
- Simplify analysis
- Compute expected utility easily
- Build the von Neumann-Morgenstern framework

Continuity Axiom

Suppose

$$A \succ B \succ C$$

Then there exists some probability

$$p \in (0, 1)$$

such that

$$B \sim pA + (1 - p)C$$

Interpretation

A middle outcome can be replaced by a suitable lottery between the best and worst outcomes.

This ensures there are no "jumps" in preferences.

Continuity: Example

Suppose

$$100 \succ 50 \succ 0$$

Continuity implies that there exists

$$p$$

such that

$$50 \sim \begin{pmatrix} 100 & 0 \\ p & 1-p \end{pmatrix}$$

The decision maker is indifferent between receiving Rs.50 for certain and a suitable gamble.

Independence Axiom

If

$$A \succ B$$

then for every lottery C and every probability p ,

$$pA + (1 - p)C \succ pB + (1 - p)C$$

Interpretation

Adding the same background risk to two alternatives should not reverse preferences.

Independence: Example

Suppose

$$100 \succ 50$$

Then Independence requires

$$0.8(100) + 0.2(0) \succ 0.8(50) + 0.2(0)$$

Since both lotteries contain the same 20% chance of zero, the ranking should remain unchanged.

von Neumann–Morgenstern Theorem

Theorem

If preferences over lotteries satisfy:

- ① Completeness
- ② Transitivity
- ③ Continuity
- ④ Independence

then there exists a utility function

$$u : X \rightarrow \mathbb{R}$$

such that preferences can be represented by expected utility.

Expected Utility

Consider

$$L = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}$$

Expected Utility is defined as

$$EU(L) = \sum_{i=1}^n p_i u(x_i)$$

Decision Rule

Choose the lottery with the highest expected utility.

Expected Utility: Example

Consider

$$u(x) = \sqrt{x}$$

and lottery

$$L = \begin{pmatrix} 100 & 0 \\ 0.5 & 0.5 \end{pmatrix}$$

Expected Utility:

$$EU(L) = 0.5u(100) + 0.5u(0)$$

$$EU(L) = 0.5(10) + 0 = 5$$

Expected Value vs Expected Utility

Lottery	EV	EU
(100,0;0.5,0.5)	50	5
(45;1)	45	6.71

Expected Value suggests choosing the gamble.

Expected Utility suggests choosing the certain payoff.

Key Insight

Individuals maximize utility, not necessarily money.

Certainty Equivalent

The Certainty Equivalent (CE) of a lottery is the amount of money that makes the individual indifferent between:

- receiving the amount for certain
- participating in the lottery

Formally,

$$u(CE) = EU(L)$$

Risk Premium

Risk Premium is defined as

$$RP = EV - CE$$

Quantity	Value
Expected Value	50
Certainty Equivalent	25
Risk Premium	25

A positive risk premium indicates risk aversion.

Why Do People Value Lotteries Differently?

Consider the lottery

$$L = \begin{pmatrix} 100 & 0 \\ 0.5 & 0.5 \end{pmatrix}$$

with

$$EV = 50$$

Different individuals may assign different certainty equivalents.

Question

Why are some people willing to avoid risk while others seek it?

The answer lies in the shape of their utility functions.

Utility Curvature and Risk Attitudes

Risk attitude depends on the curvature of the utility function.

Utility Shape	Curvature	Risk Attitude
Concave	$u''(x) < 0$	Risk Averse
Linear	$u''(x) = 0$	Risk Neutral
Convex	$u''(x) > 0$	Risk Seeking

Utility curvature determines how individuals evaluate uncertainty.

Concave Utility Function

Risk-averse individuals have:

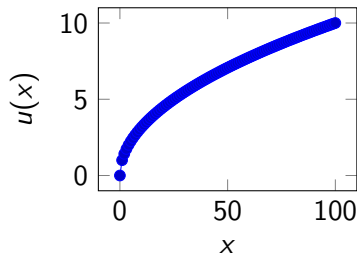
$$u''(x) < 0$$

Example:

$$u(x) = \sqrt{x}$$

Characteristics:

- Diminishing marginal utility
- Prefer certainty
- Positive risk premium



Linear Utility Function

Risk-neutral individuals have:

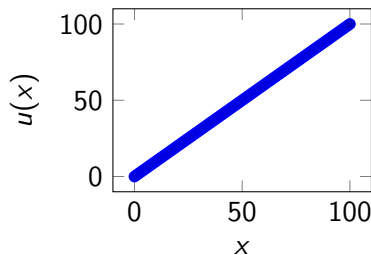
$$u''(x) = 0$$

Example:

$$u(x) = x$$

Characteristics:

- Utility proportional to wealth
- Maximizes expected value
- Risk premium equals zero



Convex Utility Function

Risk-seeking individuals have:

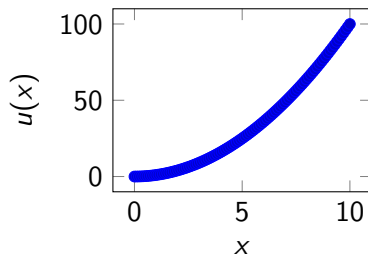
$$u''(x) > 0$$

Example:

$$u(x) = x^2$$

Characteristics:

- Increasing marginal utility
- Enjoy uncertainty
- Negative risk premium



Comparing Risk Attitudes

	Risk Averse	Neutral	Risk Seeking
Utility Shape	Concave	Linear	Convex
Certainty Equivalent	Low	Equal to EV	High
Risk Premium	Positive	Zero	Negative
Typical Behaviour	Insurance	Fair Gamble	Gambling

Why Risk-Averse Individuals Avoid Risk

For a concave utility function,

$$u(E[X]) > E[u(X)]$$

This result is known as Jensen's Inequality.

Implication

The utility of the expected payoff exceeds the expected utility of a risky payoff.

This explains risk aversion.

Insurance as Risk Aversion

Suppose:

- Wealth = Rs.100,000
- 5% chance of losing Rs.40,000

Many individuals willingly pay an insurance premium.

Why?

Reason

Reducing uncertainty may increase expected utility even if it lowers expected wealth.

Insurance is a direct consequence of risk aversion.

Beyond Risk: Decisions Across Time

Utility theory also studies decisions involving time.

Examples:

- Receive Rs.100 today
- Receive Rs.120 next year
- Invest now for future returns
- Save versus consume

To analyze such decisions we introduce discounted utility.

Why Do We Discount the Future?

Consider the following choices:

- Rs.100 today or Rs.100 next year?
- Consumption today or saving for the future?
- Immediate reward or delayed reward?

Most individuals prefer receiving benefits sooner rather than later.

Time Preference

Future payoffs are generally valued less than present payoffs.

Present Value

Suppose an amount V_t is received after t periods.

Its present value is

$$PV = \frac{V_t}{(1 + r)^t}$$

where

- r = discount rate
- t = number of periods

The larger the discount rate, the lower the present value.

Present Value: Example

Suppose

$$V_t = \text{Rs. } 1000$$

received one year from now.

If

$$r = 10\%$$

then

$$PV = \frac{1000}{1.1} = 909.09$$

Thus receiving Rs.1000 next year is equivalent to receiving approximately Rs.909 today.

Discount Factors

Instead of discount rates, economists often use a discount factor

$$\delta = \frac{1}{1 + r}$$

The discounted payoff becomes

$$PV = \delta^t V_t$$

Discount Rate	Discount Factor
5%	0.952
10%	0.909
20%	0.833

Discounted Utility

Suppose utility is obtained over multiple periods.

The discounted utility is

$$U = u(c_0) + \delta u(c_1) + \delta^2 u(c_2) + \dots$$

where

$$0 < \delta < 1$$

is the discount factor.

Future utility receives less weight than present utility.

Exponential Discounting

The standard model assumes a constant discount factor.

$$U = \sum_{t=0}^{\infty} \delta^t u(c_t)$$

Properties:

- Time consistent
- Mathematically convenient
- Widely used in economics and game theory

This model forms the basis of repeated games.

Do People Always Follow Expected Utility Theory?

Expected Utility Theory assumes:

- Completeness
- Transitivity
- Continuity
- Independence

However, experimental evidence shows that real individuals often violate these axioms.

One of the most famous examples is the Allais Paradox.

The Allais Paradox

Proposed by Maurice Allais (1953).

It demonstrates a systematic violation of the Independence Axiom.

Individuals are asked to choose between two pairs of lotteries.

Their choices often contradict Expected Utility Theory.

Allais Paradox: Choice Problem 1

Lottery A	Lottery B
100% chance of \$1M	89% chance of \$1M 10% chance of \$5M 1% chance of \$0

Most people choose:

$$A \succ B$$

because of the certainty of winning.

Allais Paradox: Choice Problem 2

Lottery C	Lottery D
11% chance of \$1M 89% chance of \$0	10% chance of \$5M 90% chance of \$0

Most people choose:

$$D \succ C$$

because of the larger jackpot.

Violation of Independence

Observed choices:

$$A \succ B$$

and

$$D \succ C$$

These preferences cannot be explained by Expected Utility Theory.

Key Insight

Human beings place disproportionate value on certainty.

This is known as the Certainty Effect.

Lessons from the Allais Paradox

The Allais Paradox shows that:

- People do not always behave according to Expected Utility Theory.
- Independence is often violated.
- Psychological factors influence decisions.
- Behavioral Economics emerged partly from such observations.

Expected Utility remains a powerful benchmark despite its limitations.

The Prisoner's Dilemma

Two players simultaneously choose:

Cooperate (C) or Defect (D)

	C	D
C	(3,3)	(0,5)
D	(5,0)	(1,1)

Payoffs are shown as

(Player 1, Player 2)

Dominant Strategy

For Player 1:

If Player 2 cooperates:

$$5 > 3$$

so Defect is better.

If Player 2 defects:

$$1 > 0$$

so Defect is still better.

Conclusion

Defect strictly dominates Cooperate.

The same reasoning applies to Player 2.

Nash Equilibrium

Since both players choose Defect,

$$(D, D)$$

is the Nash Equilibrium.

Resulting payoff: (1,1)

However

(3,3) would make both players better off.

Paradox

Individual rationality leads to collective inefficiency.

Why Repetition Changes Everything

Suppose the game is played repeatedly.

Players now care about:

- Current payoffs
- Future payoffs
- Reputation
- Future punishment

Future consequences can influence present behavior.

This creates the possibility of cooperation.

Finite Repetition

Suppose the game is played exactly T times.

Consider the final period.

There is no future punishment.

Therefore: D is optimal.

By backward induction, D is optimal in every period.

Backward Induction

Last Round



Defect

Second Last Round



Defect

Third Last Round



Defect

Infinite Repetition

Now suppose the game has no known final period.

Players maximize

$$U = \sum_{t=0}^{\infty} \delta^t u_t$$

where

$$0 < \delta < 1$$

is the discount factor.

Future payoffs become important.

Grim Trigger Strategy

Strategy:

- ① Cooperate initially.
- ② Continue cooperating as long as everyone cooperates.
- ③ If anyone defects, defect forever.

Punishment

A single deviation triggers permanent punishment.

Sustaining Cooperation

Payoff from cooperation forever:

$$3 + 3\delta + 3\delta^2 + \dots = \frac{3}{1 - \delta}$$

Payoff from defecting once:

$$5 + \delta + \delta^2 + \dots = 5 + \frac{\delta}{1 - \delta}$$

Cooperation is sustainable if

$$\frac{3}{1 - \delta} \geq 5 + \frac{\delta}{1 - \delta}$$

Tit-for-Tat Strategy

Rules:

- 1 Cooperate in the first round.
- 2 Copy the opponent's previous action.

Properties:

- Nice
- Retaliatory
- Forgiving
- Easy to understand

Tit-for-Tat performed remarkably well in Axelrod's tournaments.

One-Shot vs Repeated Prisoner's Dilemma

	One-Shot	Repeated
Future Interaction	No	Yes
Punishment Possible	No	Yes
Cooperation Stable	No	Sometimes
Role of δ	None	Crucial

Repetition fundamentally changes incentives.

Key Takeaways

- ➊ Rational preferences satisfy: Reflexivity, Completeness and Transitivity.
- ➋ Utility functions represent preferences.
- ➌ Lotteries model uncertainty.
- ➍ Expected Utility Theory provides a rule for ranking lotteries.
- ➎ Utility curvature determines risk attitudes.
- ➏ Discounting captures intertemporal choice.
- ➐ The Allais Paradox reveals limitations of EUT.
- ➑ Repeated interaction can sustain cooperation.